



Dipartimento di Elettronica e Informazione

Politecnico di Milano

20133 Milano (Italia)
Piazza Leonardo da Vinci, 32
Tel. (+39) 02-2399.3400
Fax (+39) 02-2399.3411

Advanced Computer Architecture

June 29, 2022

Prof. Donatella Sciuto and Marco Santambrogio

Name
Last Name
Professor:

You have 90' to complete the exam

Problem 1 (6pts)	
Problem 2 (6pts)	
Problem 3 (6pts)	
Question 1 (5pts)	
Question 2 (5pts)	
Question 3 (5pts)	
Total (32pts)	

To pass the exam (the following conditions are in AND):

- The overall score has to be equal or higher than 18
- At least 1 problem and 1 Question has to be correct

Problem 1

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.

(R0 is set to 100, R1 is set to 0)

```
LOOP:    LD      F3  0   R0
         ADDD    F1  F3  F3
         MULTD   F2  F3  F1
         ADDI    R1  R1  1000
LOOP2:   LD      F3  0   R1
         MULTD   F2  F2  F3
         SUBI    R1  R1  10
         BNEZ    R1  LOOP2
         SUBI    R0  R0  10
         BNE     R0  R1  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

LOOP 10 times, LOOP2 steady 100

Worst case:

1BHT, not colliding initialized with NT and NT. 1+1 misprediction for LOOP and $(1+1) * 10$ for LOOP2. Total of 22 miss.

2BHT, NT strong and NT strong. 2 + 1 misprediction@LOOP and 2 + $(1)*10$ @LOOP2. Total of 15 miss.

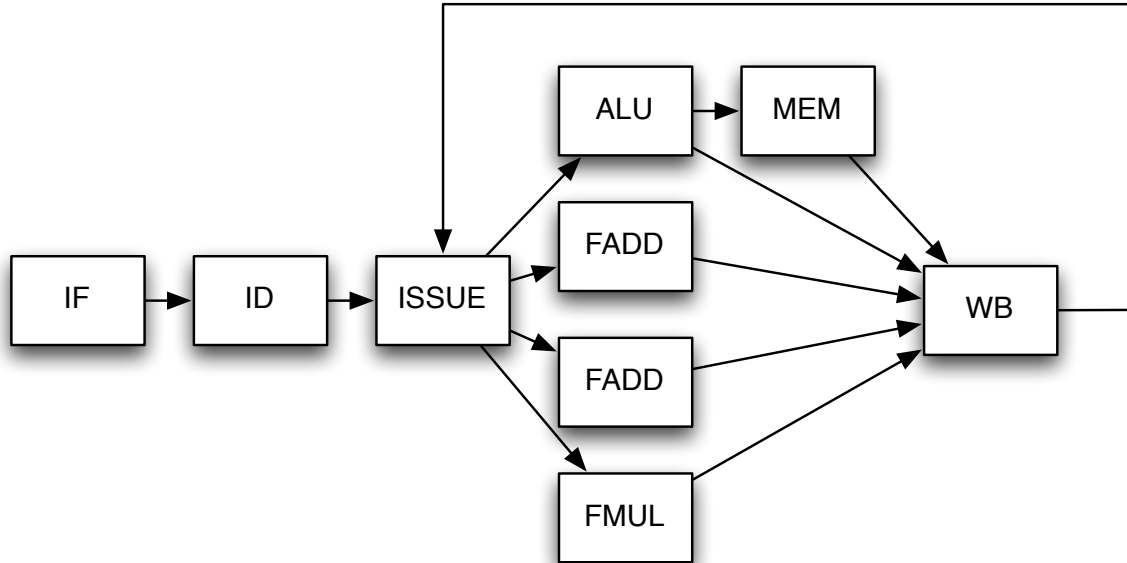
Best case:

1BHT. T and T: 1@LOOP and 1 + $2*9$ @LOOP2. Total of 20 miss.

Best of 1BHT is worst than worst-case of 2BHT

Problem 2

In this problem we will examine the execution of a code segment on the following single-issue out-of-order processor:



You can assume that:

- All functional units are pipelined
- ALU operations take 1 cycle
- Memory operations take 3 cycles (includes time in ALU)
- Floating-point add instructions take 3 cycles
- Floating-point multiply instructions take 5 cycles
- There is no register renaming
- Instructions are fetched, decoded and issued in order
- The issue stage is a buffer of unlimited length that holds instructions waiting to start execution
- An instruction will only enter the issue stage if it does not cause a WAR or WAW hazard
- Only one instruction can be issued at a time, and in the case multiple instructions are ready, the oldest one will go first

To simplify book-keeping for this problem we will only track instructions during a few critical stages:

- I - When an instruction enters the issue stage
- E - When an instruction starts execution (enters FU)
- C - When an instruction completes execution (enters WB, results available this cycle)

For this problem we will be using the following code:

```

I1    L.D   F2, 0(R1)
I2    ADD.D F1, F1, F2
I3    MUL.D F4, F3, F3
I4    ADDI  R1, R1, 8
I5    L.D   F2, 0(R1)
I6    ADD.D F1, F2, F4

```

Fill in the following table to indicate how the given code will execute. Assume all register values are available at the start of execution.

Cycle \ Inst	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
I ₁																				
I ₂																				
I ₃																				
I ₄																				
I ₅																				
I ₆																				

Answer 2

Cycle \ Inst	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
I ₁	I	E			C															
I ₂		I			E			C												
I ₃			I	E					C											
I ₄				I	E	C														
I ₅								I	E			C								
I ₆									I			E			C					

Problem 3

Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, write-allocate, and invalidation of other caches on write (instead of updating the value in the other caches).

Time	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 0	Exclusive (0)	Invalid (1)	Yes	Yes
1	P2: write block 1	Exclusive (0)	Mod (1)	Yes	No
2	P1: write block 0	Mod (0)	Mod (1)	No	No
3	P2: read block 1	Mod (0)	Mod (1)	No	No
4	P1: read block 1	Shared (1)	Shared (1)	Yes	Yes
5	P2 : write block 1	Invalid (1)	Mod (1)	Yes	No
6	P1: read block 0	Ex (0)	Mod (1)	Yes	No
7	P2: write block 1	Ex (0)	Mod (1)	Yes	No
8	P1: read block 0	Ex (0)	Mod (1)	Yes	No
9	P2: read block 0	Shared (0)	Shared (0)	Yes	Yes

Question 1

The following 5 questions are True/False.
Each of them will count 1 point

Question 1.1

A superscalar architecture can let instructions from different threads to be executed at the same clock cycle.

Given the previous statement, confirm if it is TRUE or FALSE and **effectively support** your answer.

Circle the **right** answer: True False

Answer 1.1

TRUE! A superscalar architecture has several program counters to let instructions from different threads to be executed at the same clock cycle. This technique is also known as simultaneous multithreading.

Question 1.2

The ARM big.LITTLE architecture, considering the fact that is a multicore architecture where the cores are sharing the same ISA, cannot be considered a MIMD architecture but a SIMD one.

Circle the **right** answer: True False

Answer 1.2

FALSE! A multicore architecture with different program counters, even if the cores are sharing the same ISA, is a MIMD architecture

Question 1.3

A physically centralized memory can be used to implement a shared-memory architecture

Circle the **right** answer: True False

Answer 1.3

TRUE! Both the centralized memory and the distributed memory models can be used to implement a shared-memory architecture

Question 1.4

A VLIW Architecture has to have multiple Program Counters to load the necessary Multiple Data

Circle the **right** answer: True False

Answer 1.4

FALSE! A VLIW architecture is a SINGLE instruction architecture (the bundle is a SINGLE Very Long Instruction), therefore we need just one Program Counter

Question 1.5

In a 5-stage pipeline MIPS architecture, it is possible to forward a data twice from stage-i to stage-j.

As an example, given the following code:

```
I1 ADDD F2, F4, F6
I2 ADDD F4, F2, F26
I3 ADDD F6, F2, F4
```

F2 will be forwarder from I1 to both I2 and I3 by using the EX->EX forwarding path

Circle the **right** answer: True False

Answer 1.4

FALSE! Once data is moving from one stage to the following one, we cannot use any longer the previous stage to forward it!

Question 2

Explain the difference among fine-grained multithreading, coarse-grained multithreading and simultaneous multithreading.

Question 3

1. Explain the main differences between a VLIW architecture and a superscalar architecture, in terms of instructions scheduling, hardware complexity, and code size.
2. In real superscalar processors, some architectural limits do not allow fully exploitation of the ILP present in the program. Specify these architectural limits and specify how they impact on the potential ILP